

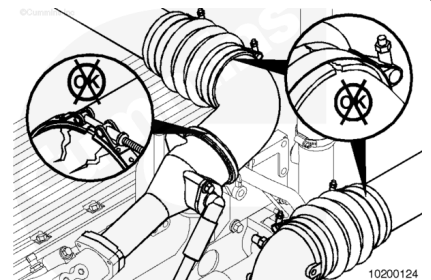
Trainings ▾

Initial Check

Automotive with CM875, CM870 and CM570

⚠ CAUTION ⚠

Engine intake air must be filtered to prevent dirt and debris from entering the engine. If intake air piping is damaged or loose, unfiltered air will enter the engine and cause premature wear.



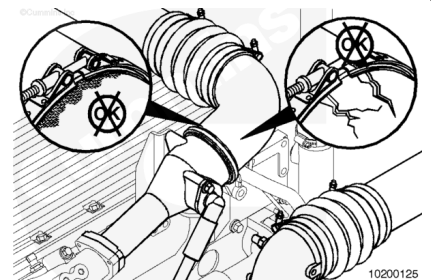
Inspect the intake air piping for cracked hoses, and damaged or loose clamps.

Replace damaged pipes and tighten loose clamps, as necessary, to make sure the air intake system does **not** leak.

Torque Value: 8 n•m [71 in-lb]

Check for corrosion of the intake system piping under the clamps and hoses. Corrosion can allow corrosive products and dirt to enter the air intake system.

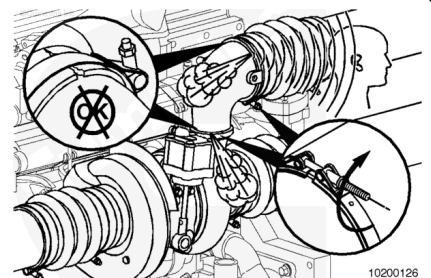
Disassemble and clean the intake air piping as required.



The noise can be caused by a turbocharger-to-discharge-elbow connection air leak.

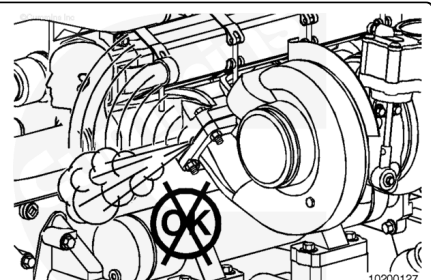
Inspect for damage. Tighten the loose clamps.

Torque Value: 8 n•m [71 in-lb]



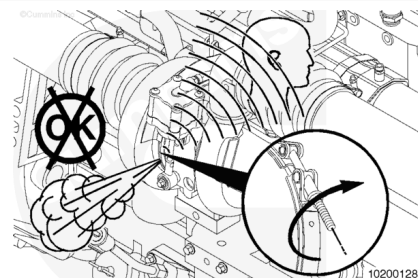
The noise can be caused by a turbocharger-to-exhaust-manifold mounting gasket air leak.

Replace the turbocharger gasket. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)



⚠ CAUTION ⚠

Do not over-torque turbine housing v-band. If v-band is overtightened it will provide less clamp load to the turbine housing and bearing housing flanges which can result in damage to turbocharger.



The noise can be caused by a turbine housing sealing surface exhaust leak.

Tighten the turbine housing v-band clamps.

Torque Value:

Capscrews

1. 14 n•m [120 in-lb]

Torque Value:

V-band

1. 11 n•m [100 in-lb]

The noise can be caused by a compressor housing sealing surface air leak.

Tighten the compressor housing v-band clamp nut.

Torque Value:

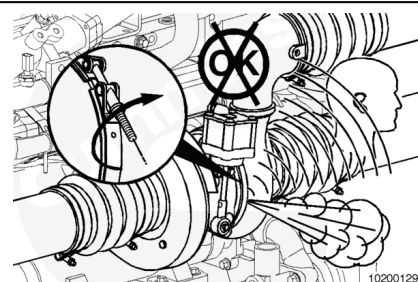
Capscrews

1. 7 n•m [62 in-lb]

Torque Value:

V-band

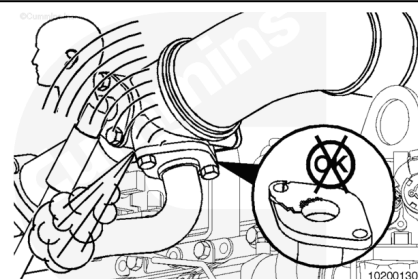
1. 9 n•m [75 in-lb]



The noise can be caused by an EGR mixer leaking exhaust from the air inlet adapter.

Inspect the air inlet adapter for cracks or other damage. Refer to Procedure 011-021 in Section 11.

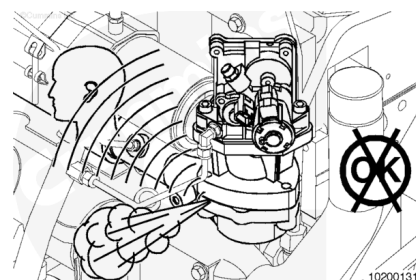
(/qs3/pubsys2/xml/en/procedures/35/35-011-021-tr.html)



The noise can be caused by an EGR valve exhaust leak at the EGR valve to exhaust manifold connection.

Check that the v-band is properly tightened. Refer to Procedure 011-022 in Section 11.

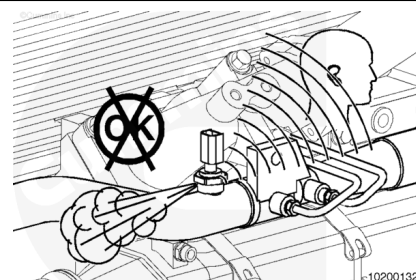
(/qs3/pubsys2/xml/en/procedures/35/35-011-022-tr.html)



The noise can be caused by an exhaust gas pressure sensor tube exhaust leak.

Inspect the exhaust gas pressure sensor tube. Refer to Procedure 011-027 in Section 11.

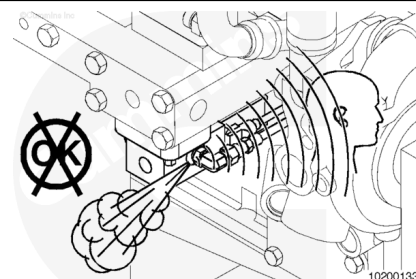
(/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html)



The noise can be caused by a turbocharger control valve air leak.

Refer to Procedure 010-119 in Section 10.

(/qs3/pubsys2/xml/en/procedures/35/35-010-119.html) Also, refer to the following procedure in the Troubleshooting and Repair Manual, CM870 Electronic Control System, ISM Engines, Bulletin 4021381. Refer to Procedure 019-388 in Section 19 (/qs3/pubsys2/xml/en/procedures/88/88-019-388.html)



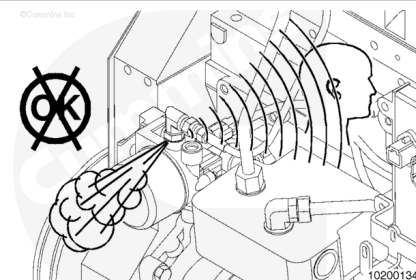
The noise can be caused by a turbocharger air shutoff valve air leak.

Refer to Procedure 010-115 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-010-115-tr.html)

Refer to Procedure 010-119 in Section 10.

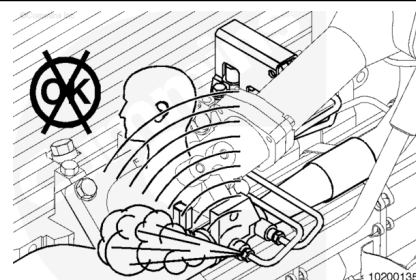
(/qs3/pubsys2/xml/en/procedures/35/35-010-119-tr.html)



The noise can be caused by an EGR differential pressure sensor tubes exhaust leak.

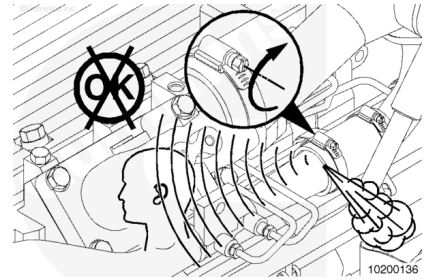
Tighten the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.

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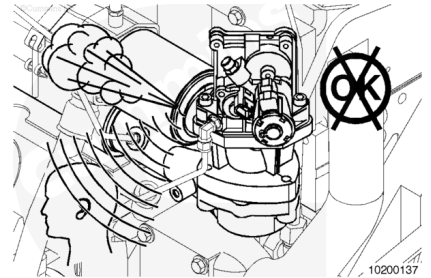
The noise can be caused by an EGR connection leak at hose connections.

Inspect the hose and check that the clamps are properly tightened.



The noise can be caused by an EGR cooler leaking exhaust from the EGR cooler connection to EGR valve.

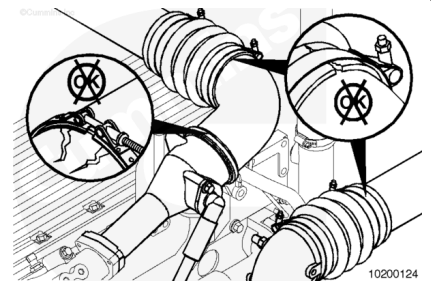
Inspect the EGR cooler connection for damage. Refer to Procedure 011-019 in Section 11.
(/qs3/pubsys2/xml/en/procedures/35/35-011-019-tr.html)



Automotive with CM876

⚠ CAUTION ⚠

Engine intake air must be filtered to prevent dirt and debris from entering the engine. If intake air piping is damaged or loose, unfiltered air will enter the engine and cause premature wear.



Inspect for loose clamps or damage between the intake air piping, air cleaner, turbocharger, charge air cooler, and intake manifold.

Replace damaged pipes and tighten loose clamps.

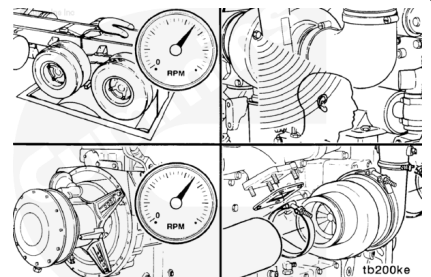
Check for corrosion of the intake system piping under the clamps and hoses.

Corrosion can allow corrosive products and dirt to enter the intake system. Disassemble and clean as required.

Excessive back pressure can cause exhaust leaks. Verify the exhaust back pressure is within specified limits. Refer to Procedure 011-009 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-009-tr.html)

Operate the engine at full throttle and maximum load, and check for air leaks. Listen for a whistling noise caused by high-pressure air leaks.

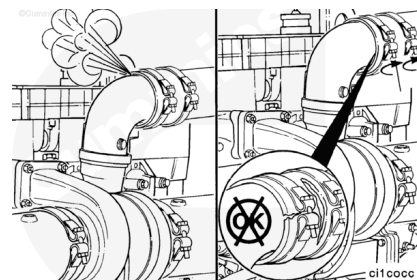


The noise can be caused by an air leak from the following:

Turbocharger-to-charge-air cooler elbow connection.

- Inspect the connection and o-ring seal for damage.
- Tighten the v-band clamps.

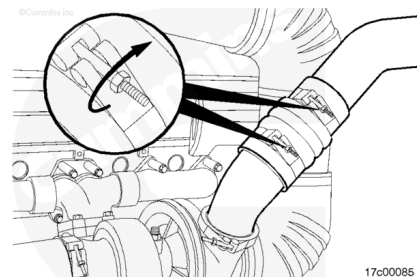
Torque Value: 9 n•m [80 in-lb]



Any charge-air cooler piping or connecting hose.

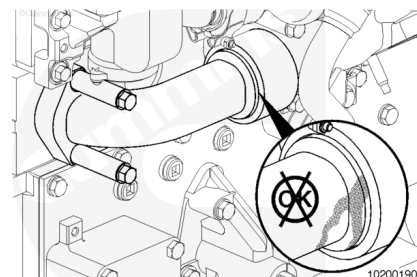
- Inspect the hose and piping for damage.
- Tighten the hose clamps.

Torque Value: 9 n•m [80 in-lb]



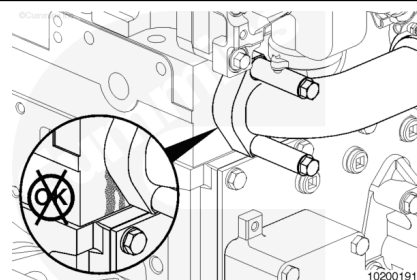
Soot streaks from the exhaust manifold slip joints.

- Remove the exhaust manifold and inspect the slip joints for damage. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html>)



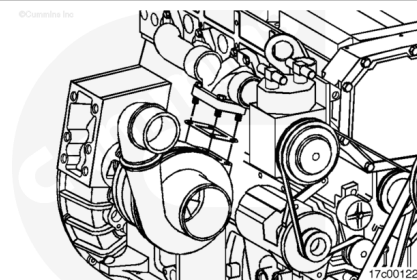
Exhaust manifold-to-cylinder head sealing gasket.

- If soot streaks are found replace the exhaust manifold to cylinder head gaskets. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-007-tr.html>)



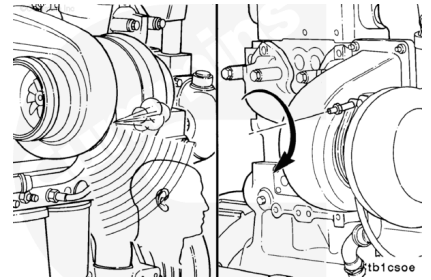
Turbocharger-to-exhaust-manifold mounting gasket.

- Replace the gasket. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)



⚠ WARNING ⚠

Removing the turbine housing v-band tamper proofing on any variable geometry turbocharger will void the Cummins warranty coverage.



Variable Geometry Turbocharger - Turbine housing sealing surface air leak.

For variable geometry turbochargers, the turbine v-band nut has a tamper proof clamp. Therefore the torque spec for the turbine v-band can **not** be checked.

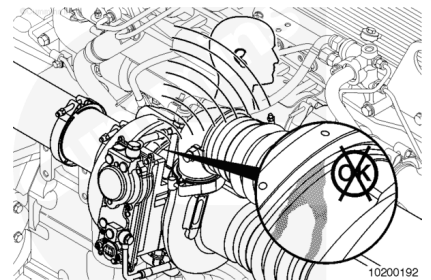
The purpose of the tamper proofing is to prevent disturbing this joint. Any loosening of the v-band or adjusting of the orientation of the turbine housing will affect the sliding nozzle clearances and will damage the turbocharger and result in poor performance.

Indications of leakage are soot streaks from the joint and/or an audible whistle from the engine. The turbocharger **must** be replaced if there are any indications of leakage from the turbine housing v-band joint.

- On non-variable geometry turbochargers, tighten the v-band clamp.

Torque Value: 14 n•m [120 in-lb]

- Check for an air leak.
- If an air leak is still present, remove and replace the turbocharger.



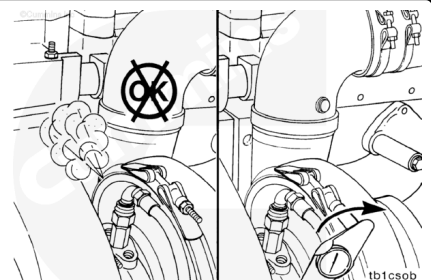
Compressor housing sealing surface air leak, all turbocharger designs.

- Tighten the v-band.

Torque Value: 9 n•m [80 in-lb]

Check for an air leak.

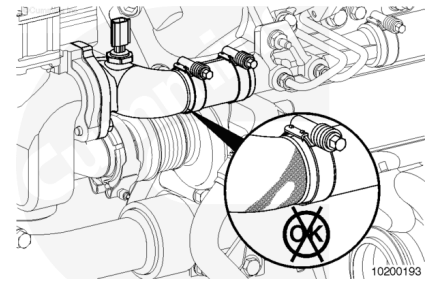
If an air leak is still present, remove and replace the turbocharger.



EGR plumbing sealing joints leak - Check for soot streaks from the EGR crossover tube and hoses.

Tighten the hose clamps. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html>)

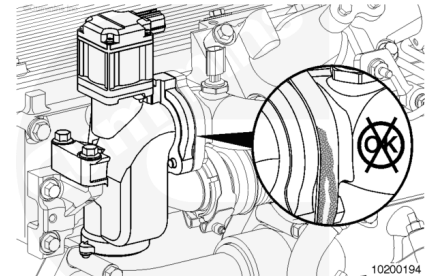
Check for an exhaust leak. If an exhaust leak is still present, remove and replace the EGR crossover tube or hose.



Check for soot streaks from the EGR valve gaskets.

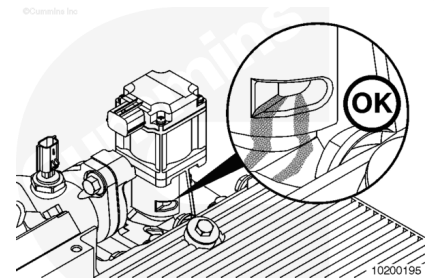
Inspect for signs of leakage such as soot streaks. If signs of leakage are present, replace the EGR valve gaskets. Refer to Procedure 011-022 in Section 11.

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Check for leakage from the EGR valve weep hole.

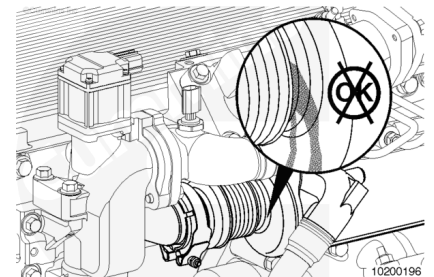
Soot leakage at the EGR valve weep hole is normal and is expected. Do **not** replace the EGR valve if soot streaks are found at the EGR valve weep hole.



Check for leakage signs (soot streaks) and for cracks.

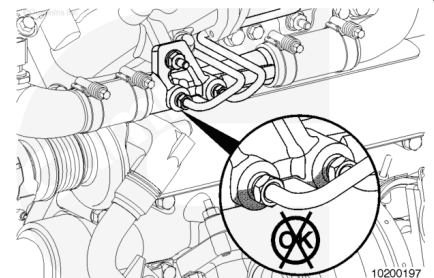
If signs of leakage or cracks are present, replace the EGR cooler or gasket. Refer to Procedure 011-019 in Section 11.

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Check for soot leakage at the EGR differential pressure sensing tubes.

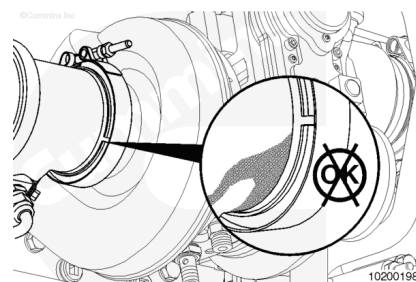
Inspect for signs of leakage such as soot streaks, which can appear yellowish brown or black. Refer to Procedure 011-026 in Section 11. (/qs3/pubsys2/xml/en/procedures/35/35-011-026-tr.html)



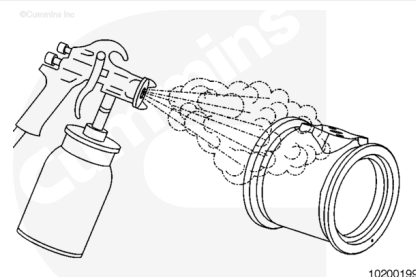
Check for soot leakage on the exhaust adapter pipe.

Inspect for signs of leakage such as soot streaks, which can appear yellowish brown or black.

If signs of leakage are present, replace the gasket and v-band clamp. See equipment manufacturer service information.



Clean all soot stains and repaint in order to detect future leaks.



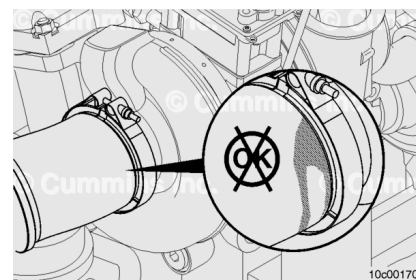
Doser Adapter Pipe - Inspect for visible soot streaks at the v-band joint.

If soot streaks are found, tighten the v-band clamp. Refer to Procedure 011-043 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html)

Operate the engine at full throttle and check for an exhaust leak.

If a leak is still present, remove and replace the gasket with a new gasket and a new v-band clamp.



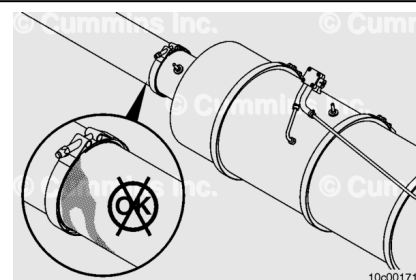
Aftertreatment Inlet Connection - Inspect for visible soot streaks at the torca clamp or v-band clamp joint. Also inspect for loose or damaged insulation wrapping (if applicable).

If soot streaks are found, tighten the v-band or torca clamp. Refer to Procedure 011-041 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

Operate the engine at full throttle and check for an exhaust leak.

If a leak is still present, remove and replace the clamp with a new torca clamp or v-band clamp.



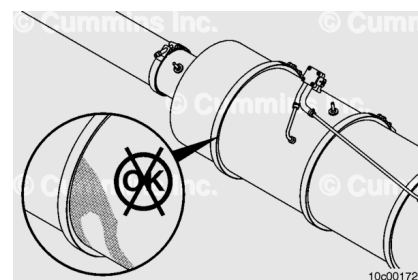
Inlet to Catalyst Connection - Inspect for visible soot streaks at the v-band joint.

If soot streaks are found, check for fault codes relating to high differential pressure in the aftertreatment system. If no fault codes are found, tighten the v-band clamp. Refer to Procedure 011-050 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-050-tr.html)

Operate the engine at full throttle and check for an exhaust leak.

If a leak is still present, remove and replace with a new gasket and v-band clamp.



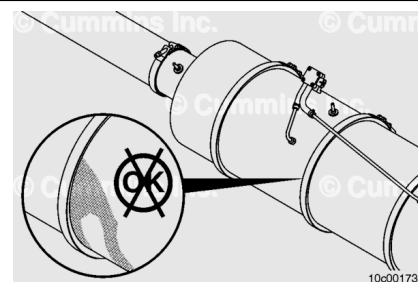
Aftertreatment Diesel Oxidation Catalyst to Aftertreatment Diesel Particulate Filter Connection - Inspect for visible soot streaks at the v-band joint.

If soot streaks are found, check for fault codes relating to high differential pressure in the aftertreatment system. If no fault codes are found, tighten the v-band clamp. Refer to Procedure 011-049 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)

Operate the engine at full throttle and check for an exhaust leak.

If a leak is still present, remove and replace with a new gasket and v-band clamp.



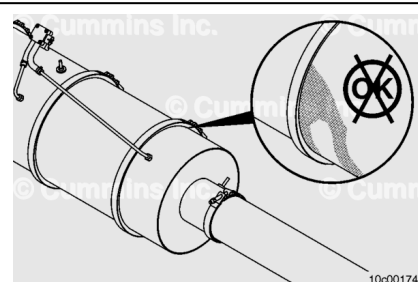
Aftertreatment Diesel Particulate Filter to Outlet Connection - Inspect for visible soot streaks at the v-band joint.

If soot streaks are found, check for fault codes relating to high differential pressure in the aftertreatment system. If no fault codes are found, tighten the v-band clamp. Refer to Procedure 011-041 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

Operate the engine at full throttle and check for an exhaust leak.

If a leak is still present, remove and replace with a new gasket and v-band clamp.



Aftertreatment Outlet Connection - Inspect for visible soot streaks at the v-band joint.

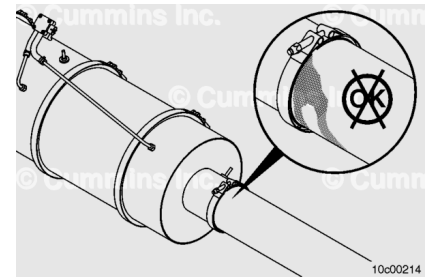
If soot streaks are found, check for fault codes relating to high differential pressure in the aftertreatment system. If no fault codes are found, tighten the v-band clamp. Refer to Procedure

011-041 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)

Operate the engine at full throttle and check for an exhaust leak.

If a leak is still present, remove and replace with a new gasket and v-band clamp.



Aftertreatment Diesel Particulate Filter Differential Pressure Sensor Tube Connections - Inspect for visible soot streaks at the threaded bosses or hose connections.

If soot streaks are found at the threaded boss, loosen the differential pressure sensor tube nut, apply a coating of anti-seize compound, and tighten the differential pressure sensor tube nut. Refer to Procedure 011-047 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html)

If soot streaks are found at the hose connections, remove and connect the hose. Refer to Procedure 011-047 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html)

Operate the engine at full throttle and check for an exhaust leak.

If a leak is still present at the threaded boss, remove and replace the differential pressure sensor tube.

If a leak is still present at the hose connection, replace the differential pressure sensor tube or differential pressure sensor, depending on which component the hose is permanently attached to.

